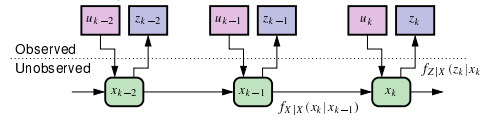




Review of probability

- Sequential probabilistic inference (and hence also any type of Kalman filter) seeks to find the best state estimate in the presence of unknown process and sensor noises.



- By definition, noise is not deterministic—it is random in some sense.
- So, to discuss the impact of noise on the system dynamics, we must understand (vector) random variables (RVs):
 - We can't predict exactly what we will get each time we measure the RV, but,
 - We can characterize the likelihood of different possible measurements by the RV's probability density function (pdf).



Review of random vectors

- As review, define random vector X , sample vector x_0 :

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}, \quad x_0 = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \text{where } X_1 \text{ through } X_n \text{ are themselves scalar RVs, and } x_1 \text{ through } x_n \text{ are scalar constant values these RVs can take on.}$$

- Random vector X is described by (scalar function) joint pdf $f_X(x)$ of vector x :
 - $f_X(x_0)$ means $f_X(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$.
 - $f_X(x_0) dx_1 dx_2 \dots dx_n$ is the probability that X is between x_0 and $x_0 + dx$.
 - $f_X(x_0)$ is the scaled probability or likelihood of measuring sample vector x_0 .



Key properties of joint pdf of random vector

- $f_X(x) \geq 0 \quad \forall \quad x$.
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_X(x) dx_1 dx_2 \dots dx_n = 1$.
- $\bar{x} = \mathbb{E}[X] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} x f_X(x) dx_1 dx_2 \dots dx_n$.
- Correlation matrix (note *outer* product, not inner product):

$$\Sigma_X = \mathbb{E}[XX^T] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} xx^T f_X(x) dx_1 dx_2 \dots dx_n$$

- Covariance matrix: Define $\tilde{X} = X - \bar{x}$. Then,

$$\Sigma_{\tilde{X}} = \mathbb{E}[(X - \bar{x})(X - \bar{x})^T] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} (x - \bar{x})(x - \bar{x})^T f_X(x) dx_1 dx_2 \dots dx_n$$



Properties of correlation and covariance

- $\Sigma_{\tilde{X}}$ is symmetric and positive-semi-definite (psd). This means:

$$y^T \Sigma_{\tilde{X}} y \geq 0 \quad \forall \quad y.$$

- Notice that correlation = covariance for zero-mean random vectors.
- The covariance entries have specific meaning:

$$(\Sigma_{\tilde{X}})_{ii} = \sigma_{X_i}^2$$

$$(\Sigma_{\tilde{X}})_{ij} = \rho_{ij} \sigma_{X_i} \sigma_{X_j} = (\Sigma_{\tilde{X}})_{ji}.$$

- The diagonal entries are the variances of each vector component
- Correlation coefficient ρ_{ij} is a measure of linear dependence between X_i and X_j ; $|\rho_{ij}| \leq 1$

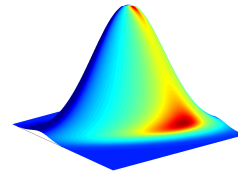


The multivariable Gaussian pdf

- There are infinite variety in pdfs; however, we assume only (multivariable) Gaussian pdf in (linear) KF.
- All noises and the state vector itself are assumed to be Gaussian random vectors.
- Gaussian or normal pdf is (we say " $X \sim \mathcal{N}(\bar{x}, \Sigma_{\tilde{X}})$ "):
 - Contours of constant $f_X(x)$ are hyper-ellipsoids, centered at $\bar{x}$, rotated via $\Sigma_{\tilde{X}}$.
 - Good news... We won't need to work directly with this equation very much!

$$f_X(x) = \frac{1}{(2\pi)^{n/2} |\Sigma_{\tilde{X}}|^{1/2}} \exp \left(-\frac{1}{2} (x - \bar{x})^T \Sigma_{\tilde{X}}^{-1} (x - \bar{x}) \right).$$

$$|\Sigma_{\tilde{X}}| = \det(\Sigma_{\tilde{X}}), \quad \Sigma_{\tilde{X}}^{-1} \text{ requires positive-definite } \Sigma_{\tilde{X}}$$



Summary

- To develop sequential-probabilistic-inference solution, must have a background understanding of random variables (RVs).
- RVs are described by probability density functions (pdfs).
- These pdfs have important properties, which we have reviewed.
- In particular, we will be making use of mean and covariance *a lot*.
- The pdf we will assume for all RVs is multivariable Gaussian (or normal) distribution.
- While this seems complicated at first, it turns out to simplify the math a lot later on.